

Harsh Shinde

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Education

KDK College of Engineering, Nagpur (RTMNU) <i>B.Tech in Electronics and Telecommunication Engineering</i>	<i>Sept 2021 – May 2025</i> <i>CGPA: 7.30</i>
Shahu's Garden High School and Jr College, Nagpur (MSBSHSE) <i>Senior Secondary</i>	<i>2021</i> <i>76.00%</i>
Shahu's Garden Convent, Nagpur (MSBSHSE) <i>Matriculation</i>	<i>2019</i> <i>67.80%</i>

Experience

Google Summer of Code <i>OSGeo</i>	<i>Remote, India</i> <i>May 2025 – Sep 2025</i>
- Introducing Geocroissant as an extension to make geospatial AI-ready metadata generation, Data-Centric AI workflows generation, validation, and conversion - Implemented multi-standard metadata conversions supporting STAC, GeoDCAT, NASA-UMM, CEDA-UK, and OGC-TDML for seamless geospatial dataset interoperability - Developed <code>mlcroissant[geo]</code> extension in croissant enabling geospatial data support in machine learning pipelines with enhanced conversion utilities	
Research Intern <i>MRSAC</i>	<i>Nagpur, India</i> <i>Jan 2025 – May 2025</i>
- Analyzed land use and land cover (LULC) changes over a two-year period, including Urban Heat Island (UHI) effects and Land Surface Temperature (LST) variations in the Nagpur region using LANDSAT-8/9 data - Applied YOLO-based models to segment building footprints from high-resolution satellite imagery in selected regions of Turkey - Performed farm boundary detection using deep learning and GIS techniques on data	

Key Projects

DeepSlide: Deep Learning-Based Landslide Detection Across Multi-Source Satellite Data	Github 🔗
- Evaluated 13 state-of-the-art deep learning models (ResNet34, VGG16, DeepLabV3+, ResNeXt50, EfficientNet, U-Net, etc.), achieving optimal performance with ResNet34 (F1: 0.747, Precision: 0.774, Recall: 0.727) - Developed a Streamlit-based interactive application for landslide detection and segmentation from satellite imagery	

Publications and Preprints

Landslide Detection and Mapping Using Deep Learning Across Multi-Source Satellite Data and Geographic Regions	Paper 🔗
Dr. Rahul Burange, Harsh Shinde , Omkar Mutyalwar Journal of Emerging Technologies and Innovative Research (JETIR), 2025	
Seasonal and Spatial Assessment of the Urban Heat Island Effect and Land Surface Temperature in Nagpur Using Landsat Remote Sensing	Preprint 🔗
Harsh Shinde , Dr. Sanjay V. Balamwar, Sanskar Shete EarthArXiv Preprint (Submitted to International Journal of Engineering and Geosciences), 2025	

Positions of Responsibility

Google Cloud Arcade Facilitator

Remote

Mentor

July 2024 – March 2025

- Mentored and guided 420+ students in Google Cloud Platform (GCP), covering Kubernetes, Cloud Functions, Apigee, Firebase, and core services like compute, storage, and networking
- Provided hands-on support with Kubernetes cluster management, serverless Cloud Functions, API management via Apigee, and Firebase integration for scalable applications
- Organized events, solved technical doubts, and maintained student engagement through continuous support and motivation

Technologies

Operating System: Windows, Linux (Ubuntu, Kubuntu)

Programming Languages: C, C++, Python, R, Bash

Machine Learning: TensorFlow, PyTorch

Database/Cache: SQL, MongoDB, Redis

Cloud: Docker, GCP

Other Software: QGIS, LaTeX, MATLAB, ESRI ArcGIS Desktop, Git, Earth Engine

Coursework

- Foundation of Machine Learning & Deep Learning
- Cloud Computing
- Advanced Satellite Image Processing (ongoing)

Achievements

Geo Spatial AI Challenge

2025

Second Runner-Up

- Our team secured 3rd place for the Tree/Orchard Detection and Counting from High-Resolution Satellite Images project in the Geo Spatial AI Challenge

MLH Hackathon

2024

Runner-Up

- A competition by MLH where we presented an Inventory Management System, built with Angular, Spring Boot, and MySQL

Extra Curricular Highlights

- Participated in Football and Chess, representing the E&TC department at the Annual Sports Meet.
- Represented KDK College of Engineering (KDKCE) at Startup Expo 2025 (Atmanirbhar Bharat 2.0) held at St. Vincent Pallotti College of Engineering & Technology, Nagpur
- Participated in an Intercollegiate Technical Competition organized by the Department of Artificial Intelligence and Data Science, KDKCE, Nagpur
- Organized a Chess Competition under the IETE Student Forum (ISF), KDKCE
- Participated in Technical Quiz Quest conducted in association with the Bureau of Indian Standards (BIS), KDKCE, Nagpur